SafeHer AI

TECHNICAL ARCHITECTURE & SAFEGUARDING SPECIFICATION

Working Prototype & Safeguarding Document (Build 1)



Project Owner: South Asia Privacy (Private) Limited (SPPL)
Funding Alignment: UNICEF Femtech Ventures Pillar 3
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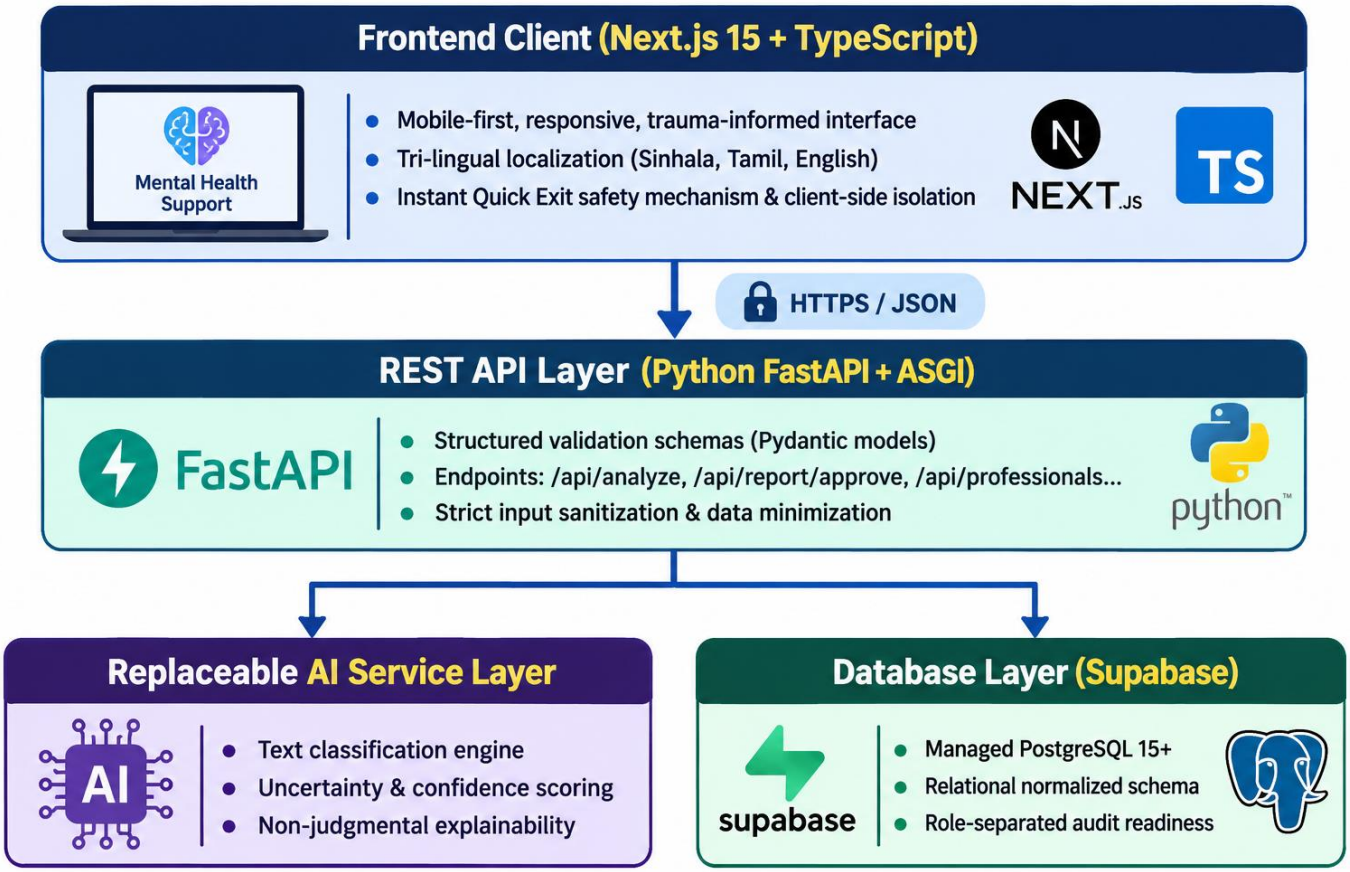
Primary Funding Alignment: UNICEF Femtech Ventures Pillar 3: Mental Wellbeing, Safety and Protection

Target Beneficiaries: Women and girls facing technology-facilitated abuse, cyber harassment, and digital threats

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1. System Architecture Overview

SafeHer AI implements a modular, privacy-first, human-centered microservice architecture designed to assist survivors while preserving strict user autonomy and data privacy.



Component Breakdown

1. **Frontend Client (Next.js 15 & Tailwind CSS):** A lightweight, progressive web client optimized for mobile-first rendering. Features zero-knowledge client isolation and a persistent Quick Exit mechanism to prevent browser snooping.
2. **Backend Engine (FastAPI):** An asynchronous Python ASGI framework handling REST operation, schema validation, and lifecycle routing.
3. **Database Architecture (PostgreSQL via Supabase):** A cloud relational database hosting core entities with strictly enforced foreign key integrity.
4. **Replaceable AI Engine:** A modular service layer capable of classifying unstructured abuse reports, producing confidence scores, and detailing the rationale behind suggested categories without relying on proprietary lock-in.

2. Core Safeguarding Principles & Human Oversight

In strict compliance with Part 3 and Part 4 of the developer specification, AI operates solely as an exploratory support tool and is prohibited from making autonomous decisions.

Strict AI Prohibitions (Non-Negotiable Boundaries)

The AI engine is architecturally and programmatically barred from executing the following actions:

- **Legal Determinations:** The system does not declare that a crime has legally occurred or evaluate the veracity of an allegation.
- **Psychological Diagnoses:** It will not diagnose mental health disorders, trauma severities, or psychiatric conditions.
- **Perpetrator Engagement:** It will never contact or notify an alleged perpetrator under any circumstances.
- **Autonomous Escalations:** The platform never executes automated calls or data dispatches to law enforcement or third parties without explicit user initiation.

Human-in-the-Loop & User Agency

- **Exploratory Framing:** Every classification is explicitly labeled as a preliminary exploratory assessment rather than a binding finding.
- **User Correction (Reclassification):** If a user feels the AI classification is inaccurate, they retain the full agency to manually reclassify the incident using a designated dropdown override.
- **Explicit Approval Gate:** Incident summaries and chronological timelines are held in staging until the survivor reviews and confirms the release checkbox, ensuring zero unsolicited sharing with professionals.

3. Separation of Adult and Child Safeguarding Pathways

To meet international child protection standards, SafeHer AI bifurcates user interactions based on age-tiering:

Parameter	Adult Pathway (18+) SafeHer AI — Working Prototype Brief and Developer Specification (1).pdf	Under-18 Pathway (Girls < 18) SafeHer AI — Working Prototype Brief and Developer Specification (1).pdf
Consent & Control	Full autonomous consent, report approval, and referral routing.	Age-appropriate UI, structured assent processes, and legal guardian involvement protocols.
Professional Directory	Standard verified legal advocates and psychosocial counselors.	Exclusively vetted, child-trained, and credential-verified child protection specialists.
Direct Messaging	Managed consultation request lifecycles.	Open, unmoderated private messaging between minors and practitioners is strictly prohibited.
Escalation Support	User-directed referrals and crisis hotline links.	Supervised safeguarding workflows involving accredited child welfare agencies.

4. Privacy, Security & Data Governance Controls

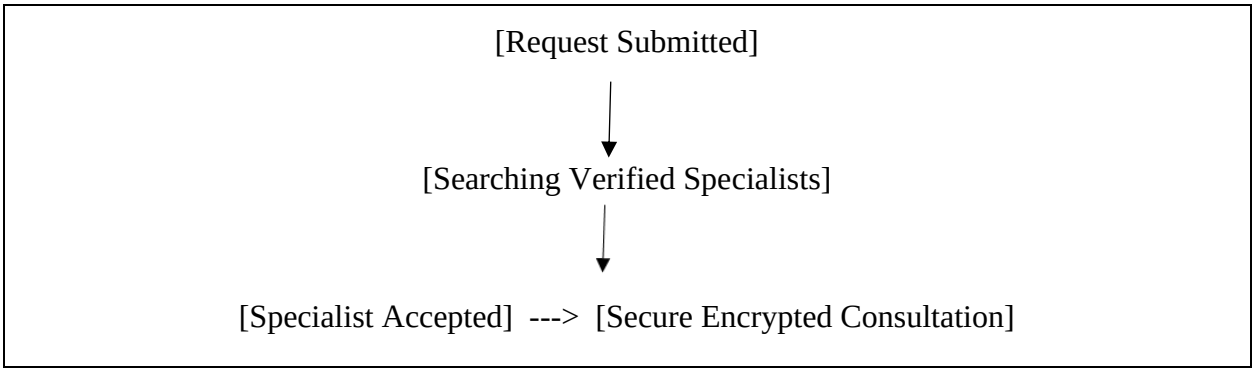
The platform implements privacy-by-design aligned with Sri Lanka's Personal Data Protection Act (PDPA) and UNICEF safeguarding standards:

1. **Data Minimization:** No personal phone numbers, national identification numbers, or real identities are required to receive an AI assessment. Records are referenced via randomized identifiers.
2. **Encryption:**
    - **In-Transit:** Enforced HTTPS / TLS 1.3 across all client-server communications.
    - **At-Rest:** Enforced AES-256 database encryption on all cloud records.
3. **No Third-Party Scraping:** SafeHer AI does not access, scrape, or monitor external social media platforms or private user device storage without active submission.

4. **Trauma-Informed UI Controls:**
    - **Quick Exit:** An immediate redirect trigger replacing the current window state with `[https://www.google.com](https://www.google.com)` to maintain user physical safety.
    - **Zero Persistent Session Artifacts:** No sensitive case details are retained in unencrypted client-side `localStorage`.
5. **Anonymized Community Impact:** The public-facing intelligence dashboard computes aggregated statistics (e.g., distribution percentages, average response times) without ever exposing identifiable survivor cases.

5. Verified Professional Marketplace & Request Lifecycle

The consultation workflow guarantees authenticated human verification before any request is processed:



- **Credential Verification Gate:** Only professionals verified and approved by system administrators can receive live consultation dispatches.
- **B2B2C Institutional Sponsorship Privacy:** Institutional sponsors can purchase consultation credits for beneficiaries, but they are barred from accessing individual case records, summaries, or identity logs.

6. Development Milestones & Compliance Checklist

- **Repository & Open-Source Licensing:** Public GitHub repo with OSI-approved MIT License.
- **Multilingual UI:** Runtime translation dictionary supporting Sinhala, Tamil, and English.
- **AI Engine & Agency:** Keyword-driven confidence assessment paired with user correction override.
- **Database Relational Integrity:** Supabase PostgreSQL deployment running normalized entity tables.
- **Human Oversight Gating:** Explicit approval checkbox required prior to professional referral generation.